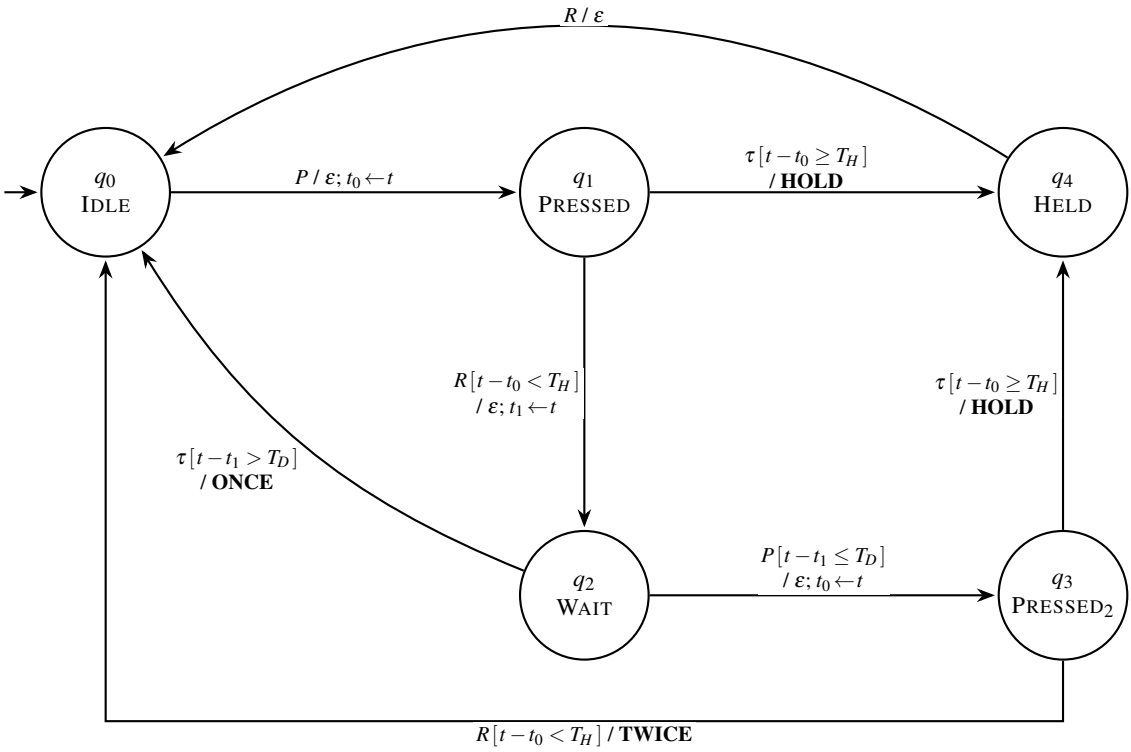


Input alphabet $\Sigma = \{P, R, \tau\}$
- P: debounced press edge (GPIO9 → LOW)
- R: debounced release edge (GPIO9 → HIGH)
- τ: clock tick (every pass of `loop()`)

Output alphabet $\Lambda = \{\textbf{ONCE}, \textbf{TWICE}, \textbf{HOLD}, \varepsilon\}$

Clocks t_0: last press, t_1: last release

Constants
- $T_{DB} = 30$ ms: an edge is accepted only if $t - t_c \geq T_{DB}$ (t_c: time of the last accepted edge)
- $T_H = 700$ ms: hold threshold
- $T_D = 350$ ms: double-press window

Notation *input* [*guard*] / *output*; *action*

Unlisted (state, input) pairs: self-loop, output ε.